

A Comparative Study of Linear Models and Neural Networks for Perturb-seq Data

1. Problem Statement and Research Questions

Understanding how gene perturbations affect cellular behavior is a central problem in computational biology. Recent advances in single-cell RNA sequencing combined with CRISPR-based perturbation technologies allow researchers to measure how gene expression changes when specific genes are knocked down. These datasets provide a unique opportunity to study causal relationships between gene perturbations and transcriptional responses.

In this project, I focus on exploring the problem of predicting gene expression responses given a gene perturbation. Specifically, I use data from the Virtual Cell Challenge[1], which contains single-cell expression profiles under different gene perturbations.

The research questions for this project are:

Can simple linear models perform competitively with nonlinear neural networks in predicting gene expression responses to perturbations in this dataset?

How sensitive are model performances to hyperparameter choices?

This project aims to provide a clear, controlled comparison between model classes using a real biological dataset.

2. Literature Review

Research on modeling cellular behavior and gene perturbation response has evolved from early mechanistic simulations to modern data-driven machine learning approaches. The following articles follow a chronological order of published date.

- Background: Perturb-seq methodology [2]
This source explains how CRISPR-based perturbations are combined with single-cell RNA sequencing to generate datasets used in this project.
- 2003: Virtual Cell (quantitative cell biology) [3]
This early work introduces the concept of “virtual cells” that simulate biological processes and emphasizes quantitative modeling for understanding cellular systems.
- 2018: Whole-cell modeling review [4]
This review discusses efforts to build comprehensive models that integrate genomic and cellular data to predict cell behavior. It highlights key challenges such as data integration, biological complexity, and computational cost.
- 2024: AI for building virtual cells [5]
This article further formalizes the idea of a virtual cell as a computational system capable of simulating cellular behavior under different conditions. It outlines how artificial intelligence can be used to construct predictive “virtual cell” models. It emphasizes that, despite large-scale omics data, current models still cannot reliably predict cellular responses to perturbations.

- 2025: Benchmarking perturbation prediction models [6]
This large-scale benchmark evaluates 27 computational methods across 29 datasets and finds that most models, including deep learning approaches, generalize poorly and often perform no better than simple baselines.
- 2025: Limitations of deep learning models [7]
This study shows that current deep learning and foundation models struggle with predicting meaningful gene expression changes, especially for unseen or combinatorial perturbations.
- 2025: Virtual Cell Challenges [8]
This paper introduces a community-driven benchmark for evaluating models that predict cellular responses. It highlights that current models fall short of capturing true biological complexity.
- 2025: Review on single-cell perturbation modeling [9]
This review discusses how machine learning is being used to predict cellular responses, emphasizing the shift toward causal modeling while noting persistent limitations in generalization.
- 2025: Review on ML for perturbation prediction[10]
This study summarizes recent advances in using machine learning with single-cell data and highlights the need for better data and models for reliable predictions.
- 2025 – Computational approaches for perturbation prediction [11]
This paper evaluates different modeling approaches and shows that models struggle to generalize across genes, cell types, and conditions.

Summary:

Across these studies, model complexity has increased significantly, but generalization remains a major challenge. Interestingly, many advanced models fail to outperform simple baselines. These findings motivate this project's focus on systematically comparing simple linear models with nonlinear neural networks to better understand how complexity is necessary.

3. Gap discussion

Despite rapid progress, there is a clear gap between model complexity and actual performance. Some studies focus on complex deep learning architectures, but evidence shows they often do not outperform simple baselines.

There are limited systematic comparisons between linear models and neural networks. Existing work emphasizes benchmark results but provides less insight into when and why simpler models may suffice. Hyperparameter sensitivity is often underexplored, even though it can significantly affect conclusions.

In this project, I will provide a controlled, head-to-head comparison of linear vs. neural network models and evaluate the robustness to hyperparameters. I will focus on interpretive insight to understand this topic in a simpler design.

4. Data Acquisition

The dataset is provided by the ARC Institute Virtual Cell Challenge[1], which contains single-cell RNA sequencing (scRNA-seq) data from **Jurkat cells** (a human T lymphocyte cell line). The presence of a 'target_gene' metadata field indicates that this is a **Perturb-seq**

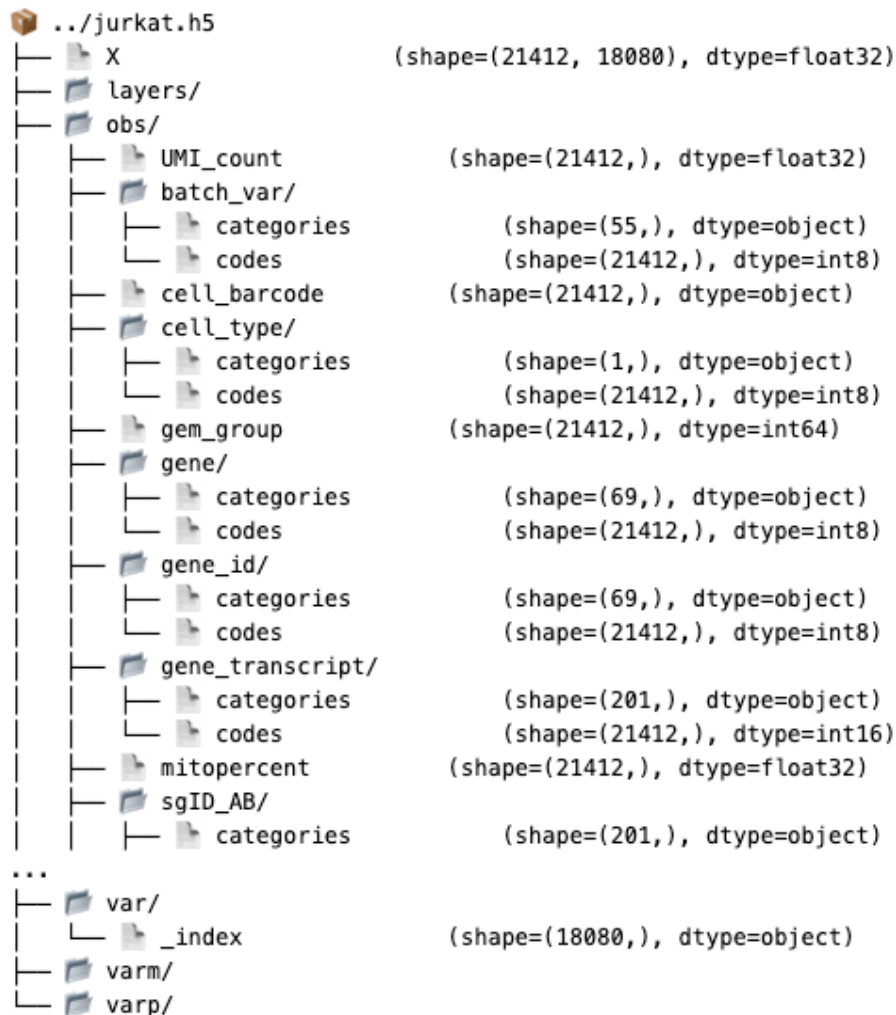
(or similar CRISPR-screen) experiment, where specific genes were knocked down to observe their effect on the cellular transcriptome.

****Total Cells:**** 21,412

****Total Genes:**** 18,080

****Format:**** HDF5

Here is a structure of the dataset made by Python:



This figure shows the hierarchical structure of the dataset in H5 format, including the gene expression matrix X, cell-level metadata obs, and gene-level annotations var. In this project, 3050 highly variable genes from the full gene set (18080) are selected.

While the dataset contains a large number of cells, the central limit theorem applies to averages of normal distribution, not to individual gene expression values. Gene expression data is typically not normally distributed due to sparsity and skewness. Log normalization helps stabilize variance and makes the data more suitable for modeling. Log Normalization was processed by the provided data.

Both the input and the target will be obtained from the “X” matrix. The control perturbation group will be used as input, and the perturbation cells will be used as the target.

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